Assignment 4

BASIC PING SWEEP WITH NMAP:

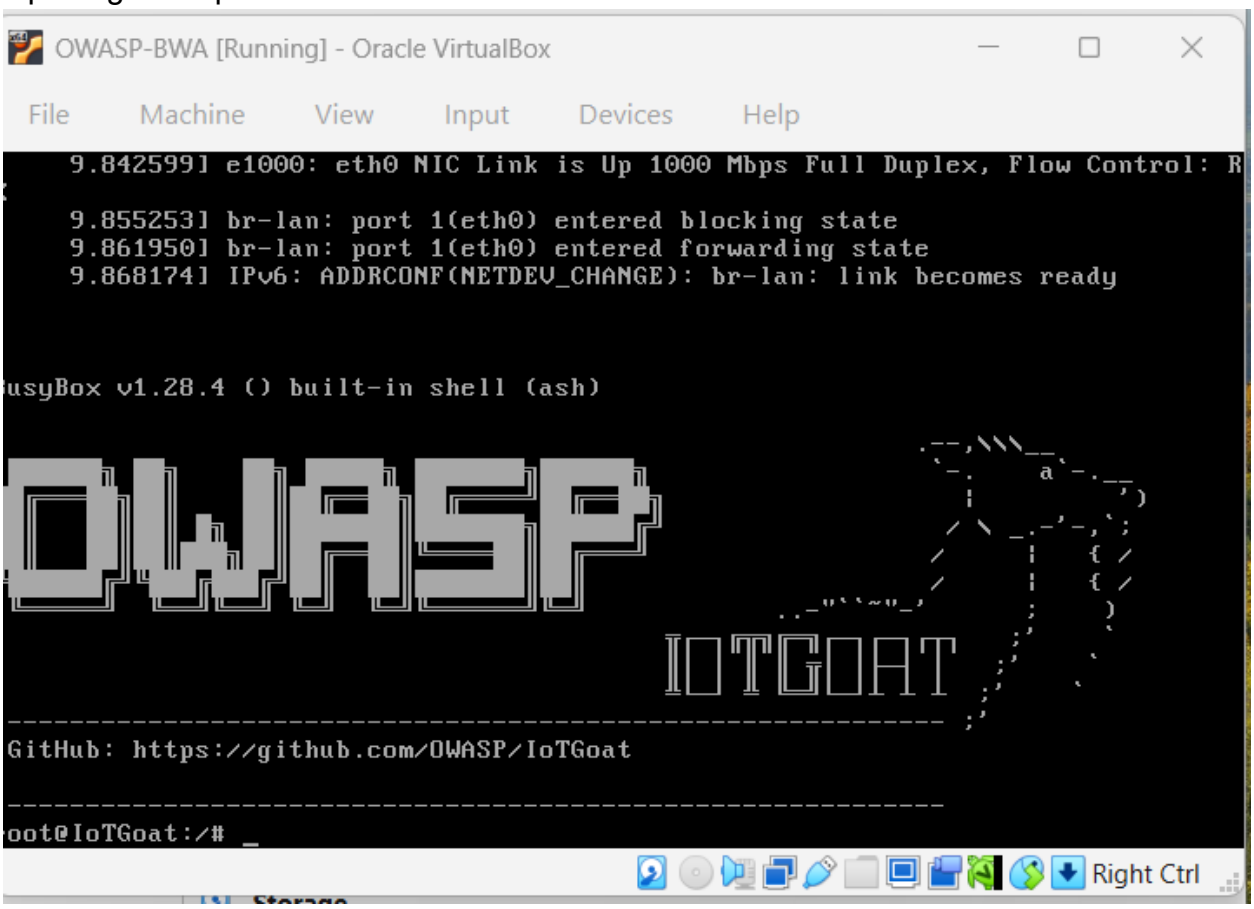
Nmap discovers alive hosts by sending different types of probe packets and analyzing the responses. During the scan, ICMP Echo Requests were sent from the scanning machine to multiple hosts within the network.

Active hosts responded with ICMP Echo Replies, confirming their availability on the network. This exchange demonstrates the standard ping-based host discovery technique.

In addition to echo requests, ICMP Timestamp Requests were also observed. These provide an alternative method of detecting live hosts, especially in environments where echo requests may be blocked by firewalls.

Overall, Nmap uses multiple techniques such as ICMP and TCP-based probes to reliably identify active systems within a network.

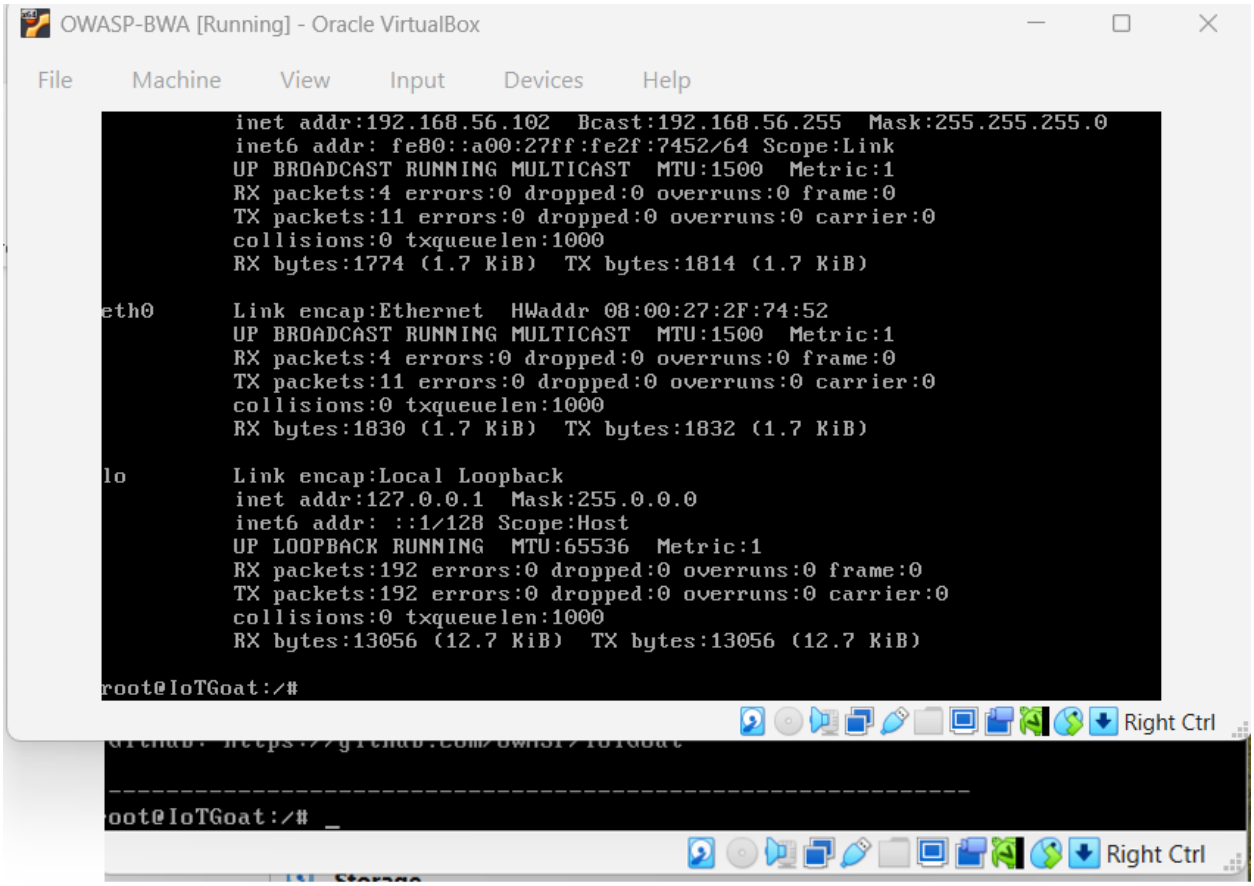
Opening Owasp:



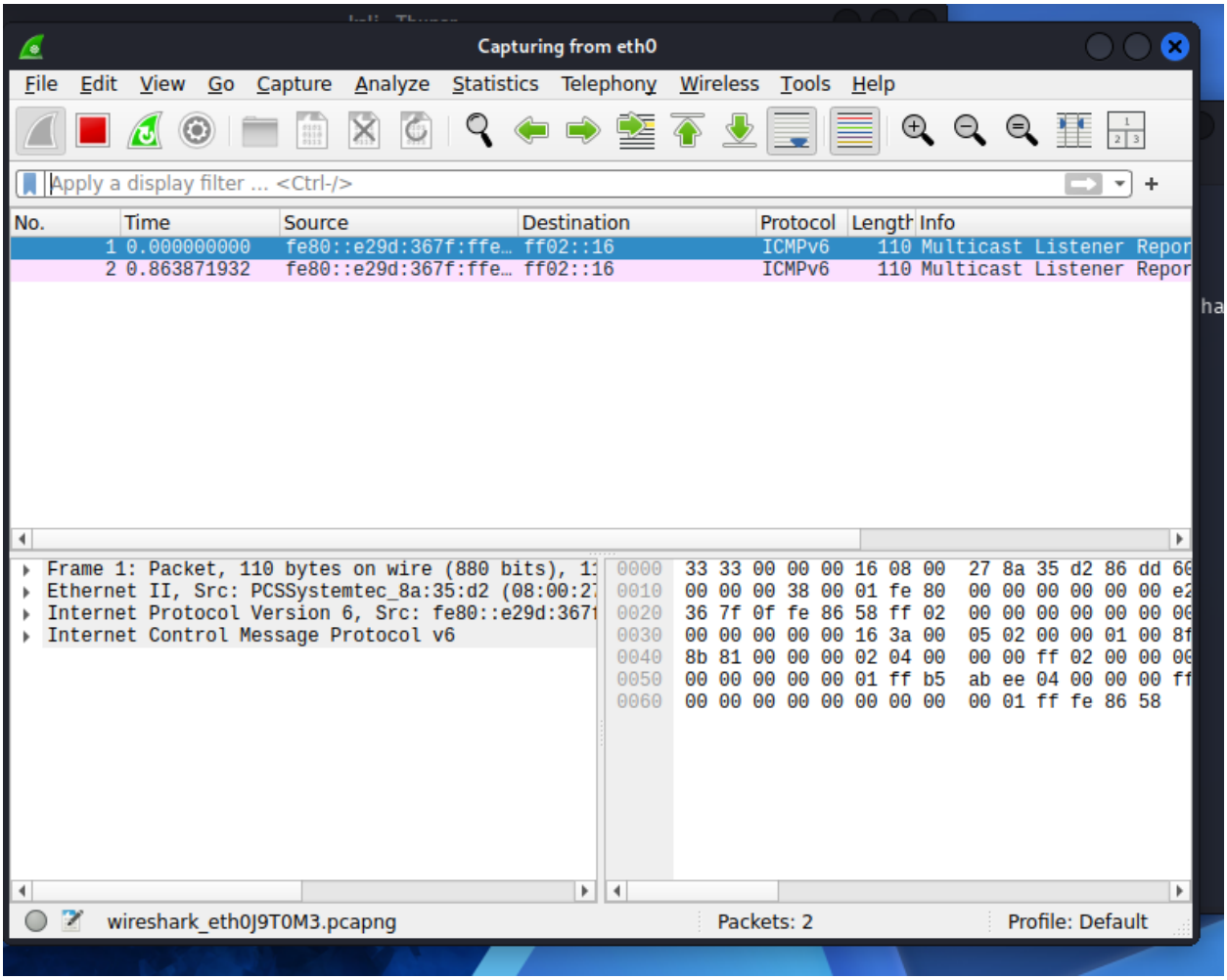
Network Details:

OWASP IP: 192.168.56.102

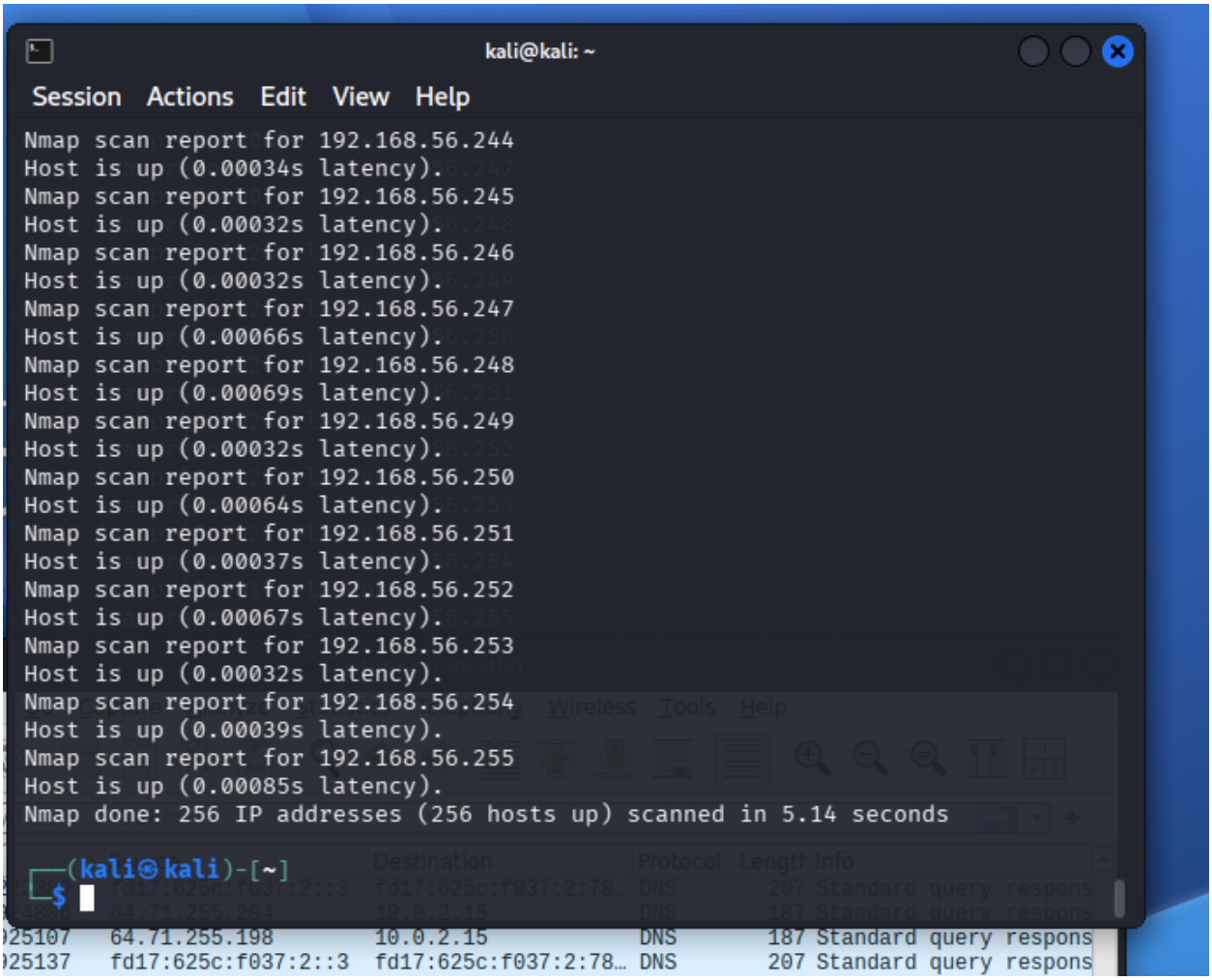
Subnet Mask: 255.255.255.0



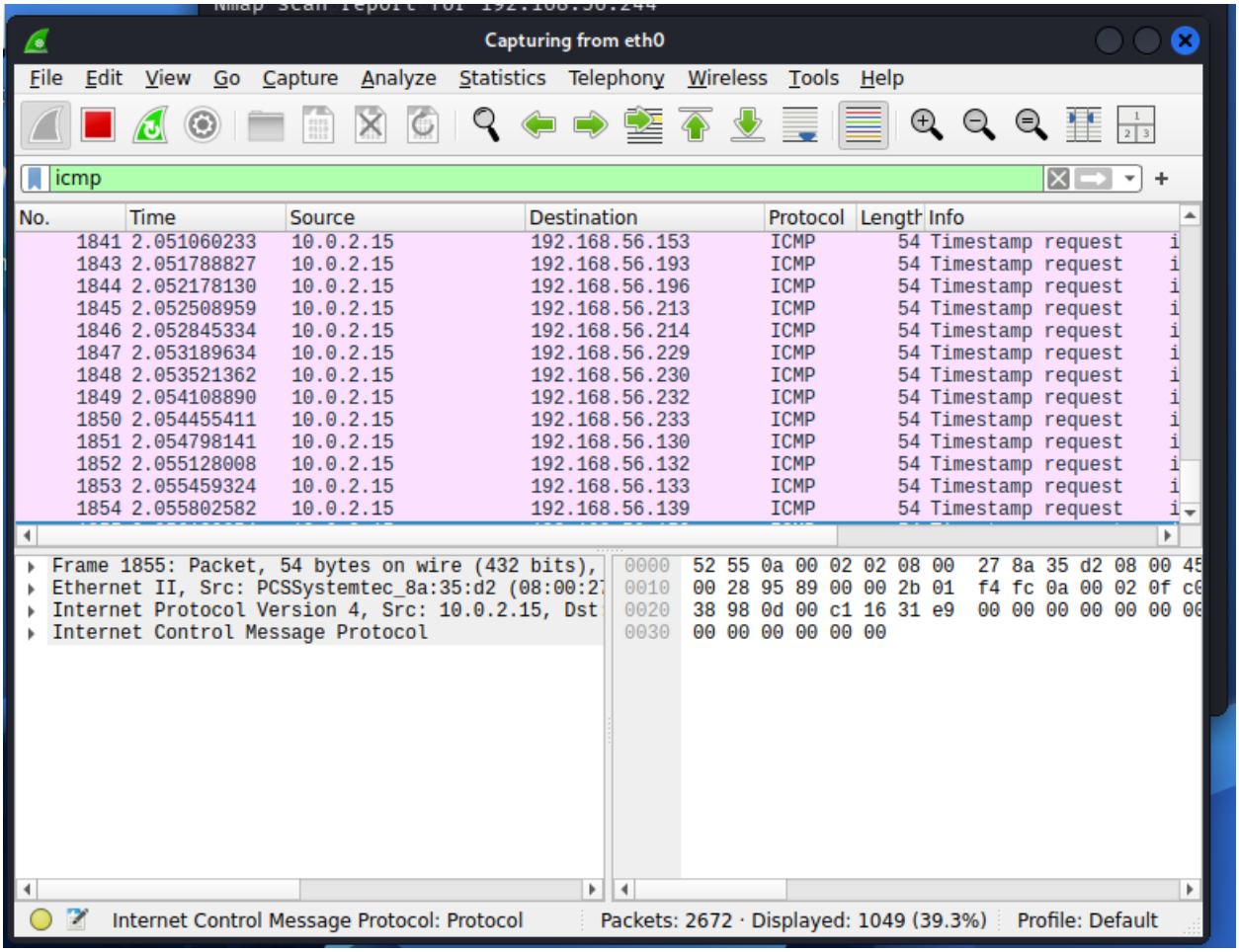
Capturing on wireshark:



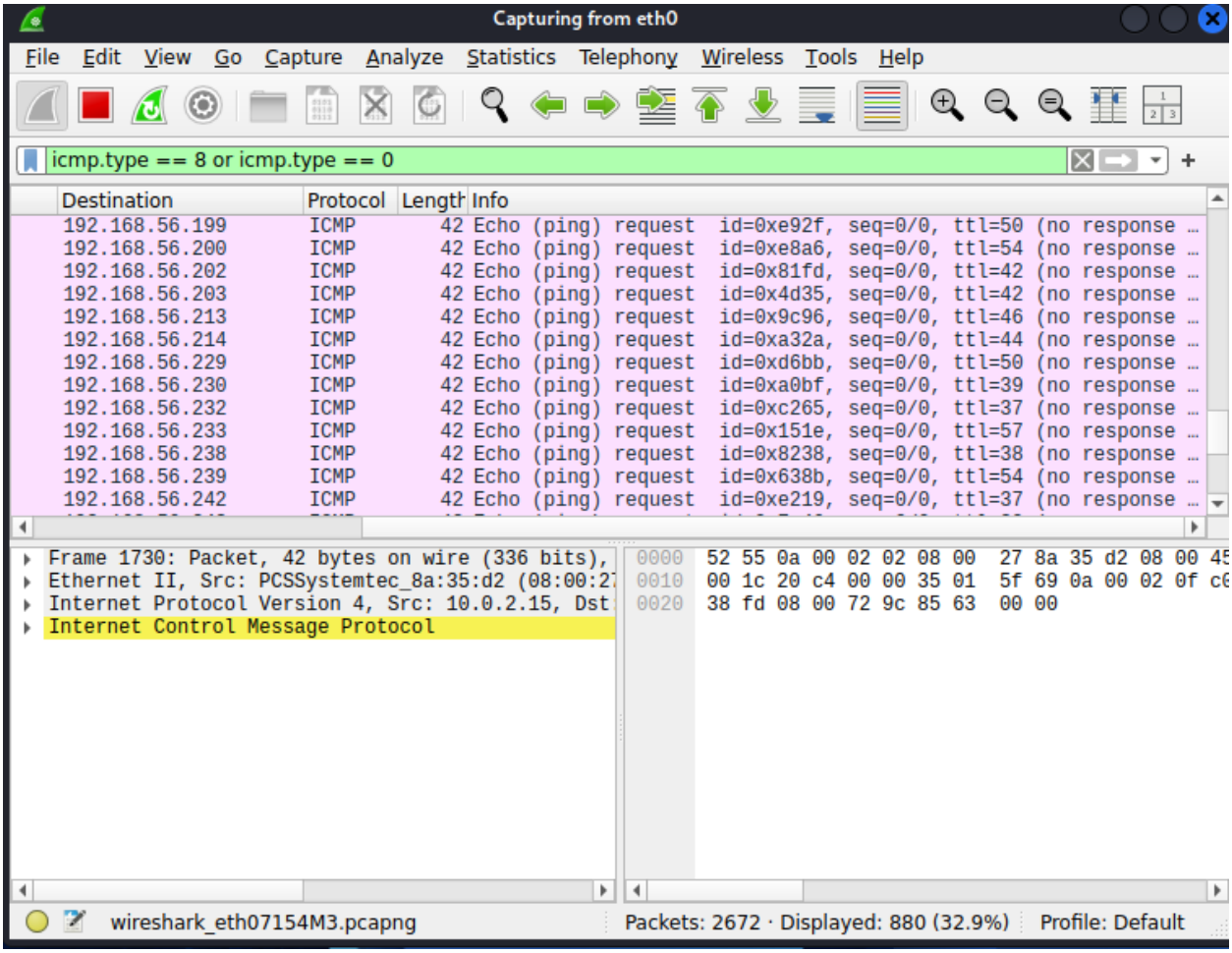
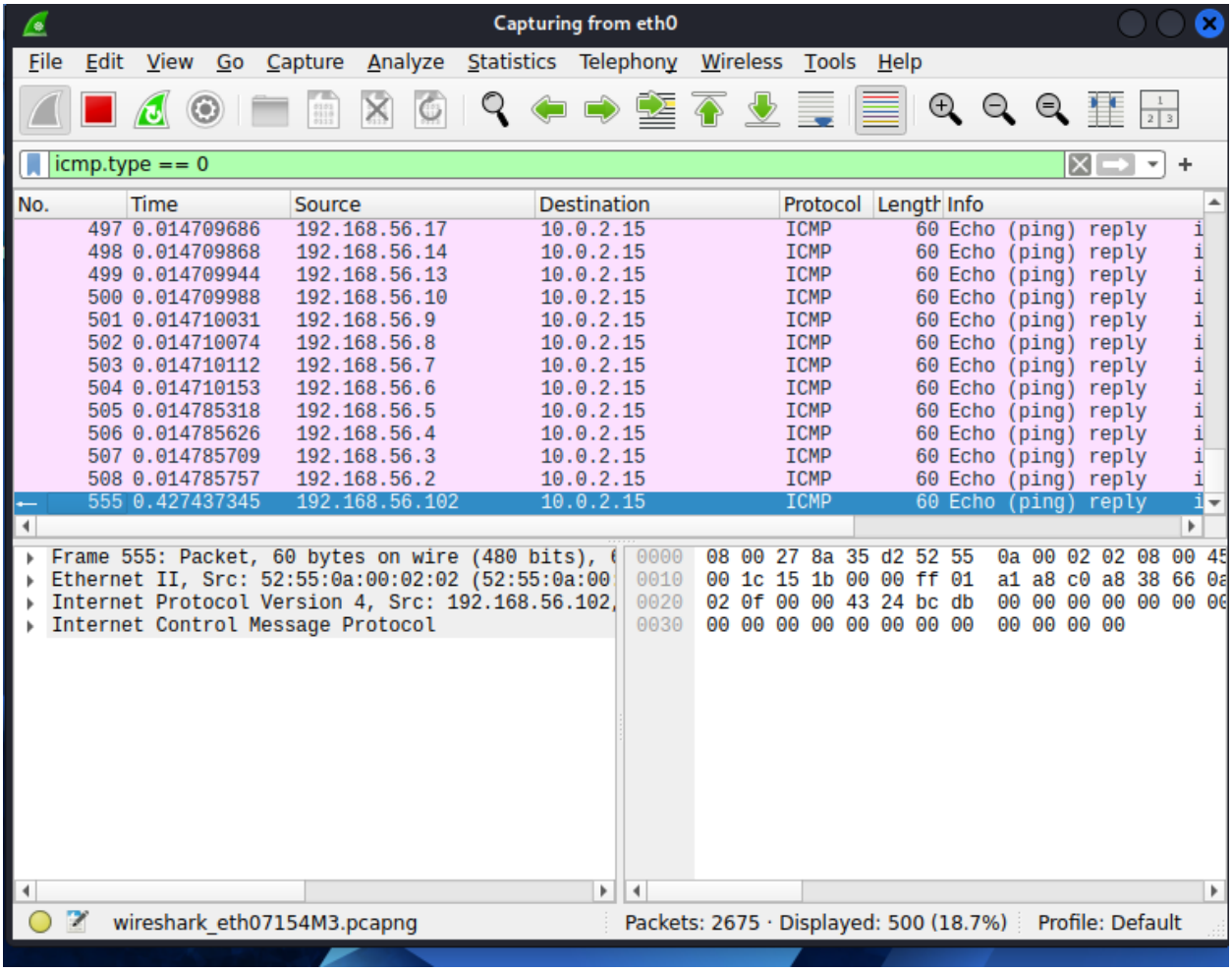
Executing nmap:



Icmp scan:


Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
1841 2.051060233 10.0.2.15 192.168.56.153 ICMP 54 Timestamp request
1843 2.051788827 10.0.2.15 192.168.56.193 ICMP 54 Timestamp request
1844 2.052178130 10.0.2.15 192.168.56.196 ICMP 54 Timestamp request
1845 2.052508959 10.0.2.15 192.168.56.213 ICMP 54 Timestamp request
1846 2.052845334 10.0.2.15 192.168.56.214 ICMP 54 Timestamp request
1847 2.053189634 10.0.2.15 192.168.56.229 ICMP 54 Timestamp request
1848 2.053521362 10.0.2.15 192.168.56.230 ICMP 54 Timestamp request
1849 2.054108890 10.0.2.15 192.168.56.232 ICMP 54 Timestamp request
1850 2.054455411 10.0.2.15 192.168.56.233 ICMP 54 Timestamp request
1851 2.054798141 10.0.2.15 192.168.56.130 ICMP 54 Timestamp request
1852 2.055128008 10.0.2.15 192.168.56.132 ICMP 54 Timestamp request
1853 2.055459324 10.0.2.15 192.168.56.133 ICMP 54 Timestamp request
1854 2.055802582 10.0.2.15 192.168.56.139 ICMP 54 Timestamp request
Frame 1855: Packet, 54 bytes on wire (432 bits),
Ethernet II, Src: PCSSystemtec_8a:35:d2 (08:00:2
Internet Protocol Version 4, Src: 10.0.2.15, Dst
Internet Control Message Protocol
0000 52 55 0a 00 02 02 08 00 27 8a 35 d2 08 00 45
0010 00 28 95 89 00 00 2b 01 f4 fc 0a 00 02 0f c0
0020 38 98 0d 00 c1 16 31 e9 00 00 00 00 00 00 00
0030 00 00 00 00 00 00
Internet Control Message Protocol: Protocol
Packets: 2672 · Displayed: 1049 (39.3%)
Profile: Default

ICMP Echo Request (type 8)



ICMP Echo reply (type 0)